DevSync

رواد مصر الرقمية

GRADUATION PROJECT

Requirements Gathering.

DevSync: Secure AI Project Collaborator
Stakeholders, stories, requirements and future ideas

TEAM OF 5

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REPORT DURATION

9 Oct 2026 - 16 Oct 2026

1 week

About this version

Requirements are unchanged in substance. This version adds an **owner** to every requirement, links each one to a feature paper, and records the new assumptions that come from the feature-ownership model. Priority labels: **Must** = protected until 30 Nov; **If time** = only after the 12 Nov core freeze; **Future** = after 30 Nov.

Required component	Where to find it
Stakeholder Analysis	Part 1
User Stories and Use Cases	Part 2
Functional Requirements	Part 3
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Traceability to the feature papers	Part 5
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Part 1: Stakeholder analysis

Stakeholder	Role	Main needs	Influence	Interest
Developer / contributor	Primary user	Projects that fit their skills; a reason for each match; an easy way to ask to join; alerts on replies	Medium	High
Project maintainer	Primary user	List a project simply; receive relevant requests; accept or decline quickly	Medium	High
Supervisor / lecturer	Assesses the project	Clear plan, working demo, evidence of requirements, tests, security, documentation	High	High
Development team (5)	Builds DevSync	Clear scope; fair workload (47 core points: Donia 12 including the finished design system, Bishay 7, Emad 8, Yousef and Saga 10); every member builds all layers of their own features; backup cover; a shippable result by 30 Nov	High	High
Pilot users	Early testers (20 or more)	Stable app, simple sign-in, relevant matches, a way to give feedback	Low	Medium
GitHub API	External service	Respect rate limits and terms; read only public data	Medium	Low
Firebase / Google Cloud	External service	Stay within quota; correct rules; billing alerts	High	Low
LLM provider	External service	Usage within budget; safe handling of prompts and keys	Medium	Low
Open-source communities (future)	Wider community	Healthy collaboration, no spam	Low	Medium

Grid. Manage closely: supervisor, development team. Keep satisfied: Firebase / Google Cloud. Keep informed: contributors, maintainers, pilot users. Monitor: GitHub API, LLM provider, communities.

How requirements are gathered and confirmed

Method	With whom	When	Output
Competitor review	Team	W1 (16-22 Oct)	Gaps and ideas (Literature Review)
Short interviews (5 developers, 2 maintainers)	Target users	W1-W2	Needs and wording for stories
Supervisor review of plan and scope	Supervisor	W1, then weekly	Approved scope and KPIs
Prototype walkthrough	Users and team	W2-W3	Usability findings (Donia's wireframes and Bishay's usability walkthrough)
Pilot with like/skip and a 5-question survey	Pilot users	Late Nov	Match quality and satisfaction data
Backlog refinement	Team	Every Monday	Updated stories and capacity check in Jira

Part 2: User stories and use cases

Personas are assumptions until interviews confirm them. **Nour** is a final-year student who knows Flutter and Python and wants a real project for her portfolio. **Karim** maintains a small open-source tool and needs two or three reliable contributors for testing and UI.

ID	Story (short)	Acceptance criteria	Priority	Feature owner
US-01	Sign in with Google or email	Both providers work; session persists after restart	Must	Donia (Authentication)
US-02	Connect GitHub account or username	Languages, repositories, contributions imported and shown	Must	Saga (Profile + GitHub data)
US-03	Add and edit my skills	Skills saved; affect the next match run	Must	Saga
US-04	Maintainer lists a project with link and skills	Project appears in the feed	Must	Bishay (Projects)
US-05	See a ranked feed with score and reason	Score and reason shown; first results in 5 s or less	Must	Emad (AI matching + feed)
US-06	Like or skip a match	Feedback saved per user and project	Must	Yousef (Like/skip)
US-07	Send a collaboration request with a message	Stored as pending; duplicates blocked	Must	Yousef (Requests + FCM)
US-08	Maintainer gets an alert on a request	Alert within 10 s in 95% of tests	Must	Yousef
US-09	Maintainer accepts or declines	Status updates for both; requester alerted	Must	Yousef
US-10	See the status of my requests	Pending, accepted, declined list	Must	Yousef
US-11	Allow or deny notification permission when needed	Asked at first use with an explanation; app works if denied	Must	Yousef (asks); Donia checks in Security rules
US-12	Smooth, clear animations	Transitions on key screens; no jank on test devices	Must	Bishay (Animations)
US-13	Sign out and delete my account data	Deletion removes profile and requests	Must	Donia (Authentication)
US-14	Install Free or Pro build from a link	Install guide works on a clean device; different app IDs and icons	Must	Saga (builds), Yousef (flavors)
US-15	Shared task board for an accepted collaboration	Create, assign, move tasks; both see updates	If time	Donia
US-16	Alerts on task changes	Alert on assignment and status change	If time	Yousef (Task alerts)
US-17	Prompt-quality dashboard	Relevance per prompt version	If time	Emad

Use cases

ID	Use case	Actors	Stories	Main owner
UC-01	Sign in	Contributor, maintainer	US-01	Donia
UC-02	Build profile from GitHub and skills	Contributor, maintainer	US-02, US-03	Saga
UC-03	List a project	Maintainer	US-04	Bishay
UC-04	Get AI-ranked matches	Contributor	US-05	Emad
UC-05	Like or skip a match	Contributor	US-06	Yousef
UC-06	Send a collaboration request	Contributor	US-07	Yousef
UC-07	Respond to a request	Maintainer	US-08, US-09	Yousef
UC-08	Track request status	Contributor, maintainer	US-10	Yousef
UC-09	Manage tasks for a collaboration (if time)	Contributor, maintainer	US-15, US-16	Donia, Yousef

UC-01 Sign in. The user taps Sign in and chooses Google or email; Firebase Auth verifies; the app creates the user document with uid and FCM token field at first sign-in, then opens the feed or profile setup. Alternatives: wrong password shows a retry error; cancelled Google sign-in returns to the sign-in screen; no internet shows an offline message.

UC-04 Get AI-ranked matches. The app sends the profile and candidate projects to the matching function, which returns a score and short reason per project; the app shows a ranked list with score meters. Alternatives: if the model is slow or fails, show skills-based ranking and a small notice; no projects shows an empty state; a fresh cached result shows immediately.

UC-06 Send a request. The user taps Request to collaborate, writes a short message, and the app stores a pending request; a Cloud Function sends an FCM alert to the maintainer. Alternatives: an existing request shows its status; if the maintainer has no token or notifications are off, the request is still stored and shown in the app.

UC-07 Respond to a request. The maintainer opens the alert or list, reads the profile, score and message, accepts or declines; the status updates and a Cloud Function alerts the contributor. If ignored it stays pending; if the contributor is offline the alert arrives on reconnect.

Part 3: Functional requirements

Each requirement is testable and has one owner, the person who builds it end to end. The Test and Security report links each FR to a test case.

ID	Requirement	Stories	Priority	Owner
FR-01	Sign in with Google using Firebase UI Auth	US-01	Must	Donia
FR-02	Register and sign in with email and password; reset password	US-01	Must	Donia
FR-03	Keep the session after restart; allow sign-out	US-01, US-13	Must	Donia
FR-04	Let a user delete their account and profile data	US-13	Must	Donia
FR-05	Read public GitHub data (languages, repositories, contributions)	US-02	Must	Saga
FR-06	Add, edit and remove skills and a short bio	US-03	Must	Saga
FR-07	Cache GitHub data and refresh in batches for rate limits	US-02	Must	Saga
FR-08	Store projects with title, description, GitHub link, needed skills	US-04	Must	Bishay
FR-09	Maintainer adds and edits only their own projects	US-04	Must	Bishay
FR-10	Include seed projects so the feed works from day one	US-05	Must	Bishay
FR-11	Build a structured profile and send it with a matching prompt through a Cloud Function	US-05	Must	Emad
FR-12	Show a fit score and short reason per project	US-05	Must	Emad
FR-13	Rank projects by score and cache per user	US-05	Must	Emad
FR-14	Fall back to skills-based ranking if the model fails or times out	US-05	Must	Emad
FR-15	Like or skip a match and store the feedback	US-06	Must	Yousef
FR-16	Keep versioned prompts and an evaluation set to compare versions	US-17	Must	Emad
FR-17	Show a prompt-quality dashboard	US-17	If time	Emad
FR-18	One collaboration request per project with a short message	US-07	Must	Yousef
FR-19	Maintainer accepts or declines a request	US-09	Must	Yousef
FR-20	Show request status to both users	US-10	Must	Yousef
FR-21	FCM alerts for new requests, status changes and project updates	US-08, US-09	Must	Yousef
FR-22	Ask notification permission at the moment of use with an explanation; work if denied	US-11	Must	Yousef (check: Donia)
FR-23	Implicit animations and advanced widgets (TweenAnimationBuilder, transitions) on key screens	US-12	Must	Bishay
FR-24	Follow the DevSync design system	US-12	Must	Donia
FR-25	Free and Pro flavors with own app ID, icon and Firebase configuration	US-14	Must	Yousef
FR-26	Obfuscated release builds and an Android App Bundle	US-14	Must	Saga
FR-27	Publish a signed build through an install link with a short guide	US-14	Must	Saga
FR-28	Two collaborators create, assign and move tasks on a shared board	US-15	If time	Donia
FR-29	FCM alerts when a task is assigned or its status changes	US-16	If time	Yousef
FR-30	Restrict the board to the two accepted collaborators with Firestore rules	US-15	If time	Donia

Part 4: Non-functional requirements

ID	Category	Requirement	Verified by	Owner
NFR-01	Performance	Match results in 5 s or less for 95% of requests	Timing logs, Performance Monitoring	Emad
NFR-02	Performance	Cold start 3 s or less on test devices	Manual timing	Yousef
NFR-03	Performance	FCM alert within 10 s in 95% of tests	Test script with timestamps	Yousef
NFR-04	Reliability	99% or more crash-free sessions	Crashlytics	Saga
NFR-05	Reliability	Backend 99.5% available during testing and pilot	Firebase status	Donia
NFR-06	Reliability	Cached data and a clear message when offline; fallback when the model fails	Offline and integration tests	Emad (fallback), Saga (tests)
NFR-07	Security	Rules let users read and write only their own data and their requests	Rules unit tests, review	Donia
NFR-08	Security	No keys in the repository; LLM key is a Cloud Function secret	Repository scan	Donia
NFR-09	Security	HTTPS only; obfuscated release builds; symbol files stored safely	Build review	Saga
NFR-10	Security	Runtime permissions only when needed, with an explanation	Manual test, checklist	Donia (check), Yousef (notifications)
NFR-11	Security	100% of the security checklist before release	Checklist review	Saga
NFR-12	Privacy	Only public GitHub data and declared skills; users can delete their data	Data review, deletion test	Saga (data), Donia (deletion)
NFR-13	Usability	Touch targets 48 dp or more; text contrast 4.5:1 (WCAG AA)	Design review, contrast check	Donia
NFR-14	Usability	New user signs in and sees matches in under 2 minutes	Walkthrough with pilot users	Bishay
NFR-15	Usability	Plain-language errors with a way to retry	Manual test	Each owner for their screens
NFR-16	Maintainability	Reviewed pull requests; 70% or more coverage of matching logic; failing tests block a pull request	GitHub, CI	Yousef (CI), Emad (matching tests)
NFR-17	Maintainability	README and technical docs updated in the same pull request as the feature	Review checklist	Each owner; Saga assembles
NFR-18	Compatibility	Two real devices and one emulator (minimum Android version to be confirmed)	Device test matrix	Saga (proposed)
NFR-19	Scalability	Pilot of 20 or more users within free quota, with budget alerts	Firebase usage	Donia

Part 5: Traceability to the feature papers

Each functional requirement sits in one feature paper; cross-cutting non-functional ones (stability, uptime, device matrix, scalability) have owners in Part 4.

Feature (owner, size, due)	Requirements covered	Needs first	Others wait on it
Foundation (Donia, 3, 22 Oct)	Data model, README, branch rules	None	Everyone
Design system and wireframes (Donia, 3, done)	FR-24, NFR-13	None	Every screen
App shell + flavors (Yousef, 3, 22 Oct)	FR-25, NFR-02	Theme (Donia)	All screens
CI and tests (Yousef, 1, 22 Oct)	NFR-16	App shell	All pull requests
Authentication (Donia, 3, 29 Oct)	FR-01 to FR-04, US-01, US-13, UC-01	Theme, app shell	Requests (user document)
Projects (Bishay, 3, 29 Oct)	FR-08 to FR-10, US-04, UC-03	Data model	Feed
Profile + GitHub (Saga, 5, 29 Oct)	FR-05 to FR-07, US-02, US-03, UC-02, NFR-12	Data model	AI matching
AI matching + feed (Emad, 5, 29 Oct / 12 Nov)	FR-11 to FR-14, US-05, UC-04, NFR-01	Seed projects, profile data	Like/skip

Feature (owner, size, due)	Requirements covered	Needs first	Others wait on it
Like/skip (Yousef, 1, 29 Oct)	FR-15, US-06, UC-05	Feed card	Prompt evaluation
Requests + FCM (Yousef, 5, 5 Nov)	FR-18 to FR-22, US-07 to US-11, UC-06 to UC-08, NFR-03	User document	Security rules, animations
Animations (Bishay, 3, 5 Nov)	FR-23, US-12	Feed, requests, profile merged	None
Prompt evaluation (Emad, 3, 12 Nov)	FR-16	Like/skip storage	Prompt dashboard
Security rules (Donia, 3, 12 Nov)	NFR-07, NFR-08, NFR-10, NFR-11 items	Request document shape	Security report
Builds + release (Saga, 3, 12 / 30 Nov)	FR-26, FR-27, US-14, NFR-09	Flavors	None
Test report + README (Saga, 2, 12 / 30 Nov)	NFR-11, NFR-17	All features	None
If time: Task board, Task alerts, Prompt dashboard	FR-28 to FR-30, FR-17, US-15 to US-17	Core freeze 12 Nov	None

Part 6: Future requirements and open questions

Outside the 30 Nov scope. Effort: S = under 1 day, M = 2-4 days, L = more than 4 days.

ID	Idea	Effort
FUT-01	Embedding pre-filter, then LLM ranking of top candidates	M
FUT-02	Issue-level matching (good first issues)	M
FUT-03	Skill-gap hints and a short learning path	M
FUT-04	Evidence in each reason (supporting repositories)	S
FUT-05	Remote Config prompt versions, A/B tests, evaluation in CI	M
FUT-06	Maintainer search for contributors	M
FUT-07	In-app chat after acceptance	L
FUT-08	Endorsements and badges	M
FUT-09	Team formation: suggest 3-4 people who fit one project	L
FUT-10	Report and block users, rate limits on requests	M
FUT-11	Firebase App Check and GitHub-verified profiles	M
FUT-12	Arabic localization with right-to-left layout	M
FUT-13	Onboarding tour and shareable profile card	S
FUT-14	Web platform on the same Firebase backend	L
FUT-15	iOS build and store release	L
FUT-16	Real Pro plan with in-app payments	L
FUT-17	Analytics dashboard for match quality, adoption and requests	M

Assumptions and open questions

#	Assumption or question	Decide by	Owner
1	Feed content is seed data plus projects added in the app (FR-08 to FR-10)	W1	Bishay
2	LLM calls run in a Cloud Function, which needs the Firebase Blaze plan	Before W3	Donia
3	The install link uses GitHub Releases or Firebase App Distribution with a signed, obfuscated build; the App Bundle is for the course	W4	Saga
4	Minimum Android version to be confirmed. LLM provider recommended: Gemini Flash-Lite through a Cloud Function, hosted open-source model as backup (see section below); budget to be confirmed	W1	Yousef (Android), Emad (LLM)
5	Stakeholder needs stay assumptions until interviews are done	W2	Donia
6 (new)	Matching input/output JSON, the user document with FCM token field, and the request document shape are agreed early so owners can build against mock data	W1-W2	Emad, Donia, Yousef
7 (new)	If-time items go to whoever finishes their core first; reviewers for them are confirmed in sprint planning	12 Nov	Donia

LLM provider recommendation

Recommendation. Use **Google Gemini Flash-Lite** (currently Gemini 3.1 Flash-Lite, the cheapest current model) for the matching score and reason, called only from a Cloud Function. Keep the model name in configuration so it can be swapped. Use a hosted open-source model (Llama, Qwen or gpt-oss through a provider such as Groq or OpenRouter) as the backup option, and do not self-host.

Option	Fit for DevSync	Cost	Main risk
Gemini 3.1 Flash-Lite (recommended)	Fast and cheap for short scoring with a JSON answer; same Google/Firebase ecosystem as the backend; free tier for development	Free tier for testing; paid $0.25 per 1M input and $1.50 per 1M output tokens (Standard); Batch about half	Free-tier content is used to improve Google products, so use the paid tier once real pilot users' GitHub data is sent
Gemini 3.5 Flash-Lite or 3.7 / 3.8 Flash	Better reasoning for tuning and comparison runs	Flash-Lite 3.5: $0.30 in, $2.50 out; Flash: $0.75 in, $3.75 out (to 31 Dec 2026, then double)	Higher cost and latency than the feed needs; use only if relevance on the evaluation set is clearly better
Hosted open-source (Llama, Qwen, gpt-oss via Groq or OpenRouter)	No vendor lock-in; very fast on some hosts; good as fallback and for the evaluation set	Per-token pricing and free tiers vary by host: confirm current prices before use	Rate limits, quality differences between models and hosts, and one more API key to protect
Self-hosted open-source	Full control over data	GPU server cost, not covered by the free Firebase quota	Setup time and cost are too high for the 30 Nov deadline; not recommended

Why this fits. The matching task is short: a profile and a few projects in, a score and one-sentence reason out. A small model is enough, and the evaluation set (target +15% relevance, 50% of top-5 rated relevant) decides whether a larger model is needed. Planning estimate: one feed run of about 2,500 input and 600 output tokens costs roughly $0.0015 on Gemini 3.1 Flash-Lite, so about 6,000 runs during the pilot cost around $9. Caching results per user and limiting calls per user keep it lower.

- Call the model only from a Cloud Function; the API key stays a function secret (or use a service account if Gemini is called through Google Cloud), never in the app or repository.
- Send only public GitHub data and declared skills (NFR-12); use the paid tier for pilot users.
- Run the evaluation set on Gemini 3.1 Flash-Lite and one open-source model in W3-W4 and keep the better one; the fallback ranking from skills stays in place whichever model is chosen.
- Owner: Emad (decide by end of W1 together with Donia for the Blaze plan and budget). Prices and model names change often, so check Google's pricing page before the final budget (last checked 9 Oct 2026).